June 2020

FORECASTING

Lagos Flood

STAGE 2 TASK

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About Lagos

Lagos, the most populous city in Nigeria and one of the fastest-growing cities worldwide, is situated in the southwest of the country. encompassing an estimated 1,171.28 square kilometers. With a population estimated to exceed 15 million as of 2023, Lagos stands as the most populous city in Africa. The city's urban classification is derived from a blend of commercial centers, industrial zones, and high-density residential areas.

Flood Occurrences In Lagos

Lagos is prone to flooding, with many floods occurring there each year, especially during the rainy season, which runs from April to October. A number of factors, such as urbanization, climate change, and inadequate infrastructure, have contributed to an increase in the frequency of floods. There have been several notable flood events in recent years that have resulted in significant property damage, transportation disruptions, and resident displacement.

Reasons behind Lagos' floods

Flooding in Lagos happens mainly because of natural and human reasons

- Heavy Rain: During the rainy season, Lagos gets a lot of rain. Its location on the coast means it often gets heavy downpours, causing sudden floods.
- Continuous Rain: When it rains for days, the ground gets too wet. Water bodies like rivers and lakes overflow, causing more floods.
- Poor Drainage: Many parts of Lagos don't have good drainage systems. This means water can't flow away properly when it rains. Blocked drains make this worse.

These factors, combined, make flooding a big problem in Lagos.

Data Preparations

Data for this study came from Visual Crossing, which gives daily details from 2013 to 2024. It covers temperature, humidity, rainfall, wind speed, wind gust, moon phase, sea level pressure, cloud cover, visibility, solar radiation, and solar energy. Flood information was collected from articles and publications focusing on recorded flood events in Lagos from 2021 to 2024.

Coding Environment: Google Collab
Tools: Python
Libraries: Pandas, Numpy, Matplotlib, Scikit-Learn, XGBoost

Description of Dataset
Weather Data:
Content: Daily weather observations from 2009 to 2024, including temperature, humidity, precipitation, wind speed, wind gust, moon phase, sea level pressure, cloud-cover, visibility, solar radiation, and solar energy.

Flood data:
Manually sourced, data found could not be trusted, gathered dates of floods from news headlines and inputted into a new sheet for cross referencing.

Cleaning and Filtering Data

The dataset was cleaned by:

1. Filtering data to the date range from January 1, 2012, to August 31, 2024.
2. Removing unwanted columns.
3. Formatting the date to the proper format.

Here is the process in code:

```
# Load data
df = pd.read_excel('/content/drive/MyDrive/Lagosflood_combined_data.xlsx')

# Convert 'date' column to datetime format
df['date'] = pd.to_datetime(df['date'])

# Define your date range for filtering
start_date = '2012-01-01'
end_date = '2024-08-31'

# Filter the DataFrame based on the date range
filtered_df = df[(df['date'] >= start_date) & (df['date'] <= end_date)]

# Drop unwanted columns
columns_to_drop = ['name', 'snow', 'snowdepth', 'uvindex', 'severerisk',
                'sunrise', 'sunset', 'conditions', 'description',
                'icon', 'stations', 'windspeedmax', 'windspeedmin']

filtered_df = filtered_df.drop(columns=columns_to_drop)

# Print the filtered DataFrame
filtered_df.head()
```

```
filtered_df.head()
```

dew	humidity	precip	precipprob	...	windgust	windspeed	winddir	sealevelpressure	cloudcover	visibility	solarradiation	solarenergy	moonphase	date
25.1	81.4	7.1	100.0	...	83.5	35.3	207.0	1009.7	75.0	7.6	NaN	NaN	0.41	2009-01-01
24.1	83.3	5.1	100.0	...	25.9	22.1	201.6	1009.1	74.2	8.6	NaN	NaN	0.50	2009-01-01
24.5	80.8	7.9	100.0	...	25.9	18.4	230.1	1008.7	64.2	9.6	NaN	NaN	0.54	2009-01-01
23.6	85.3	7.9	100.0	...	64.8	33.5	147.1	1010.1	65.2	7.1	NaN	NaN	0.58	2009-01-01
23.6	81.2	4.1	100.0	...	50.0	46.4	233.9	1008.0	75.0	9.5	NaN	NaN	0.10	2009-01-01

First 5 Rows

3272 Rows
22 Columns

Exploratory Data Analysis (EDA)

Critical understanding of the relationship and connections in the dataset was made possible by the EDA. The goal of this analysis was to identify the correlations, trends, and patterns that might affect the frequency of floods in Lagos.

	tempmax	tempmin	temp	feelslikemax	feelslikemin	\
count	3272.000000	3272.000000	3272.000000	3272.000000	3272.000000	
mean	30.749358	24.303515	27.117971	36.990556	24.687500	
min	24.700000	10.000000	23.600000	24.700000	6.800000	
25%	29.000000	23.100000	26.000000	33.600000	23.100000	
50%	31.000000	24.100000	27.000000	37.400000	24.100000	
75%	33.000000	25.000000	28.200000	40.600000	25.000000	
max	47.000000	28.900000	31.100000	50.600000	35.400000	
std	2.447803	1.839170	1.451568	4.645353	2.686944	

	feelslike	dew	humidity	precip	precipprob	...	\
count	3272.000000	3272.000000	3272.000000	3272.000000	3272.000000	...	
mean	29.933710	24.290037	85.216901	9.021638	89.700489	...	
min	23.600000	17.000000	57.500000	0.000000	0.000000	...	
25%	27.200000	23.600000	82.500000	0.300000	100.000000	...	
50%	29.600000	24.300000	85.300000	1.900000	100.000000	...	
75%	32.500000	25.000000	87.900000	7.800000	100.000000	...	
max	40.000000	27.000000	98.300000	299.000000	100.000000	...	
std	3.323334	0.941651	4.143373	20.876631	30.399895	...	

	windgust	windspeed	winddir	sealevelpressure	cloudcover	\
count	3180.000000	3166.000000	3166.000000	3231.000000	3272.000000	
mean	30.459748	23.723121	220.525490	1012.298050	60.754951	
min	0.000000	0.300000	1.000000	1007.300000	1.200000	
25%	24.100000	17.600000	207.525000	1011.000000	53.300000	
50%	28.800000	21.600000	222.700000	1012.300000	59.600000	
75%	33.500000	25.900000	237.400000	1013.600000	67.100000	
max	137.200000	222.500000	358.900000	1017.200000	100.000000	
std	12.174077	12.345362	34.603401	1.807514	11.344208	

Statistics of data

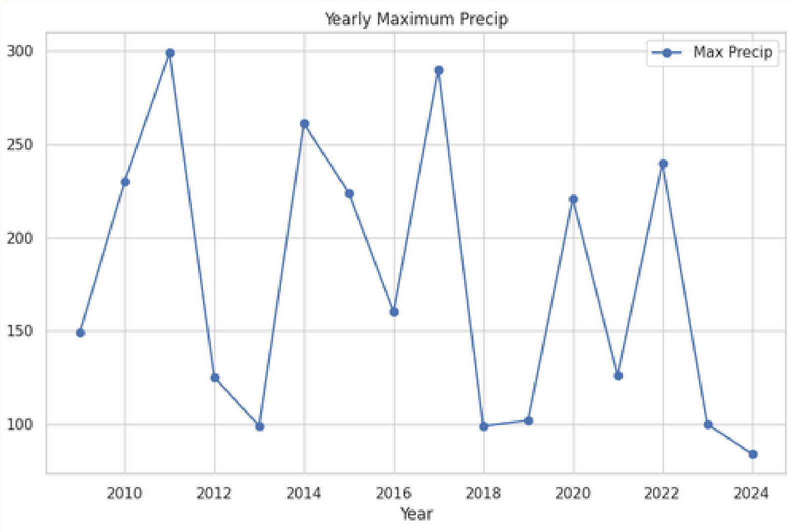
```
[8 rows x 21 columns]
<class 'pandas.core.frame.DataFrame'>
Index: 3272 entries, 66 to 6118
Data columns (total 22 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   tempmax           3272 non-null   float64
 1   tempmin           3272 non-null   float64
 2   temp              3272 non-null   float64
 3   feelslikemax      3272 non-null   float64
 4   feelslikemin      3272 non-null   float64
 5   feelslike         3272 non-null   float64
 6   dew               3272 non-null   float64
 7   humidity          3272 non-null   float64
 8   precip            3272 non-null   float64
 9   precipprob        3272 non-null   float64
 10  precipcover       3272 non-null   float64
 11  preciptype        3272 non-null   object
 12  windgust          3180 non-null   float64
 13  windspeed         3166 non-null   float64
 14  winddir           3166 non-null   float64
 15  sealevelpressure  3231 non-null   float64
 16  cloudcover        3272 non-null   float64
 17  visibility        3147 non-null   float64
 18  solarradiation    3090 non-null   float64
 19  solarenergy       3090 non-null   float64
 20  moonphase         3166 non-null   float64
 21  date              3272 non-null   datetime64[ns]
```

Total Rows 3272
Total Columns 22

Exploratory Data Analysis (EDA)

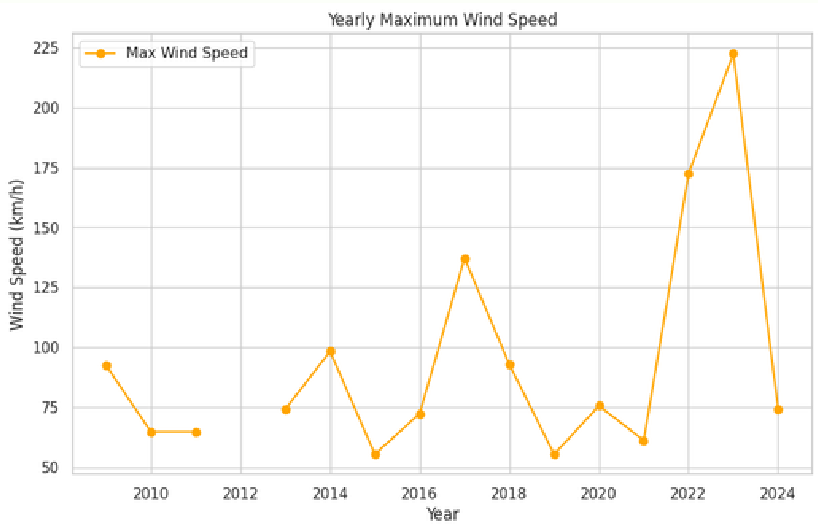
Critical understanding of the relationship and connections in the dataset was made possible by the EDA. The goal of this analysis was to identify the correlations, trends, and patterns that might affect the frequency of floods in Lagos.

1.Yearly Maximum Precipitation ,Temperature and Humidity Trends

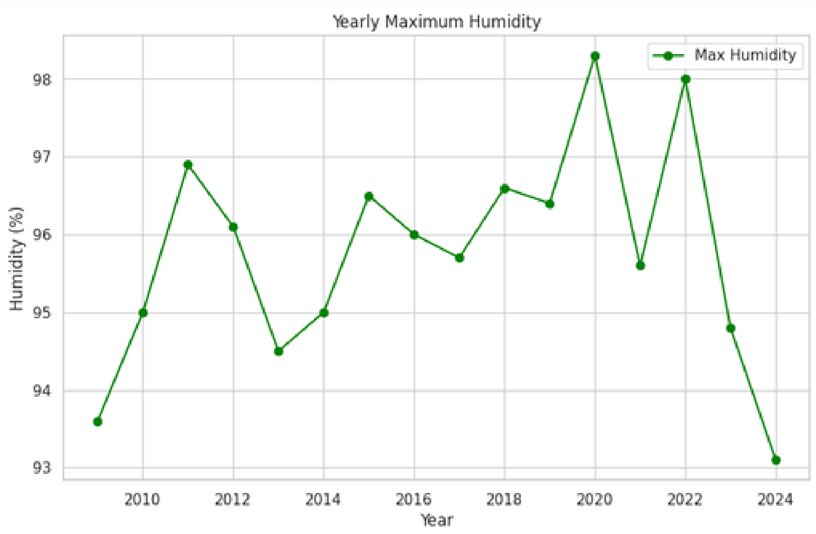


2011 experienced the highest precipitation

with a fair distribution among other years while 2024 is trending below the average

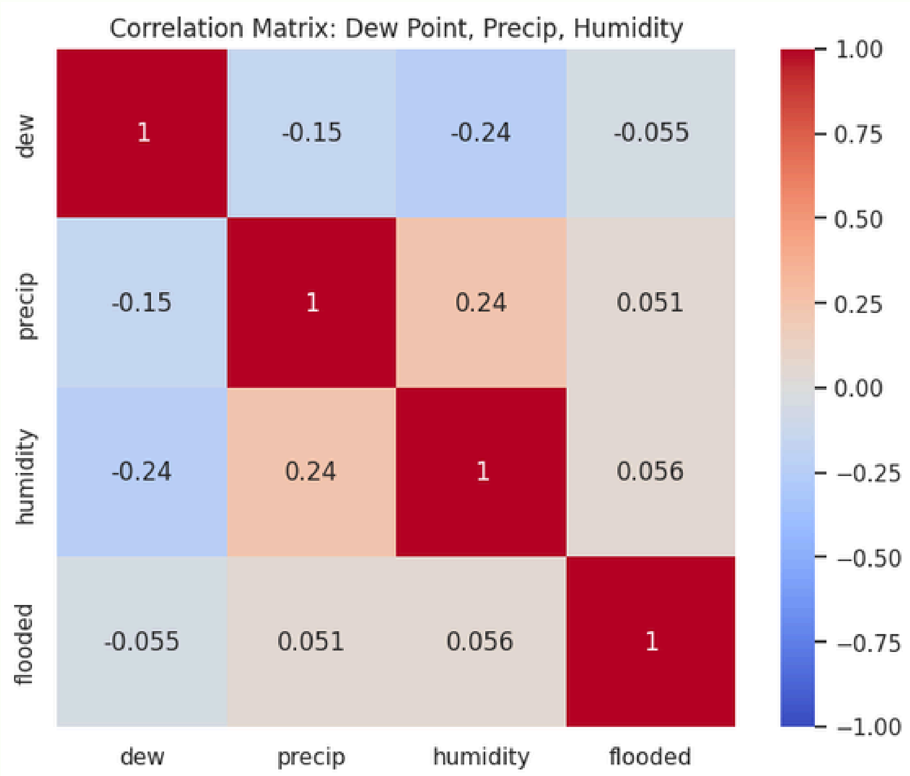


2023 experienced the highest wind speed followed by 2017



2020 recorded the highest Humidity followed by 2020

3.Correlation Analysis



1. Dew Point vs. Flooded:

- **Correlation Coefficient (-0.055): There is a very weak negative correlation between dew point and flooded events. While this correlation is weak, it suggests that higher dew points may slightly decrease the likelihood of flooded events, indicating a nuanced relationship.**

2. Precipitation vs. Flooded:

- **Correlation Coefficient (0.051): There is a very weak positive correlation between precipitation and flooded events. This indicates that higher precipitation levels may slightly increase the likelihood of flooded events, though the relationship is minimal.**

3. Humidity vs. Flooded:

- **Correlation Coefficient (0.056): There is a very weak positive correlation between humidity and flooded events. This suggests that higher humidity levels may slightly increase the likelihood of flooded events, highlighting another subtle aspect of their relationship.**

Summary:
The relationships between dew point, precipitation, humidity, and flooded events are very weak, with correlations ranging from -0.055 to 0.056. This suggests that while dew point, precipitation, and humidity may have some influence on flooded events, their impact is minimal. Other factors likely play a more significant role in causing floods.

Further investigation is necessary to understand how these variables interact with environmental and infrastructural factors to affect flooded events more comprehensively. These findings highlight the complexity of flood dynamics and emphasize the importance of considering multiple factors in flood risk assessment and management.

In conclusion, while the correlations are weak, they still provide valuable insights into understanding floods. It underscores the need for holistic approaches in managing flood risks effectively.

Modeling to Predict Flood Events

Data Preparation and Feature Engineering
The dataset underwent several preprocessing steps to prepare it for modeling. Initially, datetime information was separated from numerical data for efficient handling.

```
# Separate datetime column
date_col = filtered_df.select_dtypes(include=['datetime'])
numeric_data = filtered_df.select_dtypes(include=['number'])
```

Missing numerical values were imputed using the mean value across each column. After imputation, additional features were engineered to enhance model performance. For instance, a rolling mean of precipitation over a 3-day window, temperature differences, and interactions between humidity and temperature were calculated.

```
# Impute missing values
imputer = SimpleImputer(strategy='mean')
data_imputed = pd.DataFrame(imputer.fit_transform(numeric_data), columns=numeric_data.columns)

# Combine datetime column back with imputed data
final_data = pd.concat([date_col.reset_index(drop=True), data_imputed], axis=1)

# Feature engineering
final_data['precipitation_rolling_mean'] = final_data['precip'].rolling(window=3).mean()
final_data['temperature_diff'] = final_data['tempmax'] - final_data['tempmin']
final_data['humidity_temp_interaction'] = final_data['humidity'] * final_data['temp']
```

Remaining missing values were filled using backward filling (bfill) method to ensure completeness in the dataset.

```
# Fill any remaining NaNs after feature engineering
final_data.fillna(method='bfill', inplace=True)  # Or use another method like 'ffill' or a const
```

Data Normalization
To ensure consistency in the range of features, data normalization was performed using Min-Max scaling. This transformation mapped numerical values to a range between 0 and 1, except for the datetime column which retained its original format.

```
# Normalize data (excluding date column)
scaler = MinMaxScaler()
numerical_data = final_data.select_dtypes(include=['number'])
data_normalized = pd.DataFrame(scaler.fit_transform(numerical_data), columns=numerical_data.columns)
```

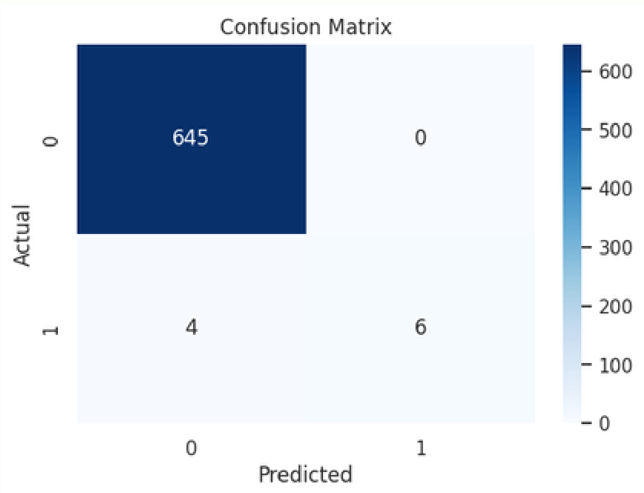
Methodology

Model Development
The target variable, 'flooded', was defined based on specific conditions related to humidity, precipitation, precipitation coverage, and wind speed. These conditions were combined into a binary classification where '1' denotes a flood event and '0' otherwise.

The dataset was split into training and testing sets (80/20 ratio) to facilitate model training and evaluation. A Linear Regression model was initially trained for comparison purposes, followed by a Random Forest Classifier.

Training and Evaluation
The Random Forest Classifier, with 100 estimators and a fixed random state, was trained on the training data. Predictions were generated and evaluated using various classification metrics:

Accuracy: Measures overall model correctness.
Precision: Indicates the proportion of true flood predictions among all positive predictions.
Recall: Reflects the proportion of true flood events correctly identified by the model.
F1 Score: Harmonic mean of precision and recall, providing a balanced measure.
ROC-AUC Score: Evaluates the model's ability to distinguish between flood and non-flood instances.
Confusion Matrix: Summarizes model predictions against actual outcomes.



Accuracy: 0.9938931297709923
Precision: 1.0
Recall: 0.6
F1 Score: 0.7499999999999999
ROC-AUC: 0.8
Confusion Matrix:
[[645   0]
 [4   6]]

Forecasting

Future Data Acquisition

Forecasted meteorological data for upcoming dates was obtained and prepared for simulation. This data included temperature, humidity, precipitation, and other relevant variables.

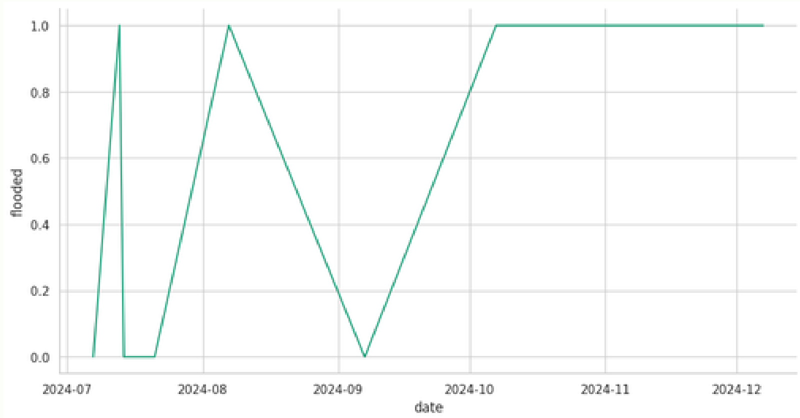
Data Processing and Prediction

The future meteorological data underwent preprocessing similar to the training phase, ensuring consistency in feature engineering and normalization.

The trained model was applied to predict flood events based on the simulated future meteorological data.

Based on the predictions from the model:

July 7th, 2024 is predicted not to experience flooding (Flooded_Predicted = 0).
July 10th, 2024 is predicted to experience flooding (Flooded_Predicted = 1).
July 11th, 2024 is also predicted to experience flooding (Flooded_Predicted = 1).



Conclusion

Summary of Findings

The simulation demonstrates the model's capability to forecast flood events based on meteorological forecasts.

Further validation and refinement against real-world data are crucial for enhancing the model's accuracy and reliability.

Link to Excel file

Link to Code file

Link to Future data file